

Almost Human

They come in all shapes and sizes and are growing more intelligent by the day. Meet the robots of today and delve into the exciting field of robotics

Samir Makwana

Pick up any newspaper today and you'll find news about a humanoid waitress at a restaurant, a dancing robot, a dog-like robot, war-bots, etc. In today's world there exists a huge scope for a career in robotics given the various applications that the field lends itself to—medical, nanotechnology and even modern warfare. Ergo, there has been a gradual rise in demand for technical experts and research specialists in robotics. No longer does robotics involve the manufacture of small automated machines, which are merely used to pick up things and perform other such simple tasks.

The Home Front

At the inauguration of a new building of the Defence Research and Development Organisation in 2006, the Prime Minister announced that India would pursue technologies for the development of a robot army. Post that, the development of war-bots is already in the pipeline for critical combat activities and espionage purposes. In January this year, the first Robo Expo 2008 was organised by the Confederation of Indian Industry (CII) after realisation that the robotics industry in India was on the path of growth. According to industry estimates, the world market for industrial robots will reach 139,300 units by 2010 with a yearly growth average of 4.2 per cent. The Indian robotics industry is expected to grow at 2-2.5 times the global average.

"The Indian robotics industry has huge potential for growth as well as employment generation. Development of low cost robots for Indian industry is one of the focus areas we are working on at present. Application development is going to be a major employment generator in this area," says Dr. T. Asokan, Assistant Professor, Dept. of Engineering Design, IIT-Madras.

Under the initiatives of DST (Department of Science and Technology) a consortium has been formed with two IITs (Madras and Bombay) and the private sector for the design and development of low-cost robots.

There are many manufacturers who utilise robotics and automation, but very few who actually manufacture robots and automation products or services "Very few industries manufacture robots in India. One has to convince labour of the necessity—more so in existing units. New units may be able to implement robotics in their plants. The people who have experience are mostly in academics or national labs," says Prof. C. Amarnath, professor at Mechanical Engineering Department, IIT Bombay.

With just a few organisations specialising in robotics, interaction with academia has become a must for training of employees in this field. "Regular interaction



3D: Shrikrishna Patkar

of companies who use or manufacture robots with the educational institutes engaged in robotic teaching or research (is a must). Institutes can do short term courses for the companies whereas the live projects can be provided to the students of academic institutes. This will keep the dialogues on," says Dr. Subir Kumar Saha, a professor at IIT-Delhi.

Enter Intelligence

Robots can be defined as machines with human-injected intelligence that are automatically controlled, can be reprogrammed, could be immobile or otherwise, and serve as multipurpose machinery. Robotics is all about designing robots, developing applications to control them, manufacture them, maintain them and conducting further research for better performance. This field overlaps with different areas like electronics, computer science, artificial intelligence, mechatronics, nanotechnology and bioengineering.

Regarding the robotics field, Asokan says, "Robotics being an interdisciplinary field requires a good amount of exposure to different areas such as mechanical design, electronics, sensors and actuators, programming, and integration of electromechanical systems. Those who plan to pursue a career in robotics need to focus on these areas and must try to develop an overall idea of integration of many systems."

According to VS Mahalingam, Director, Centre for Artificial Intelligence and Robotics (CAIR), Defence Research and Development Organisation (DRDO), "Opportunities in robotics lie in three broad areas—hardware, software and control. In hardware such as embedded systems, a single board circuit is designed and made to perform functions like other computer circuit boards (the motherboard). Next, software meant to execute complex functions involves working on the real-time issues through usage of particular Real Time Linux OS to co-ordinate with the special hardware. Critical algorithms are designed and fine tuned for the accurate performance of the software and RTOS in sync with the hardware. Finally, control of the robots is basically managing the integration of the hardware and software of the robot for achieving specified tasks."

The major areas to specialise under this field are:

Design: For designing the structure of the robotics application, specialisation in engineering design is essential. It's the initial stage of building robots wherein development engineers and even R&D engineers play a vital role in a robot's concept and design. Also the designing of hardware and embedded chips used in robots can draw



Advancements in technology in terms of memory, speed, accuracy of computers and robots and telecommunications have provided the basis for significant expansion of the field of applied robotics."

Dr. Kaustubh Chokshi
Founder and CEO
Intelligent Business Systems

influences from electrical and electronics engineering.

Implementation and Control: Software applications along with RTOS, mostly RT Linux, are used for developing critical algorithms and defining the functions to be executed by the robot through the applications of expertise from computer science and information technology. Various operational or functional issues are solved in real-time with help of application programming. The control portion includes certain aspects of electrical engineering. Also the maximum application of mechatronics takes place in this area, directly. Integration of hardware and software takes place to build a robot in this area.

Operation and Maintenance: The area involves knowledge and expertise from different engineering branches for using robots and automated machinery for achieving specified goals. The maintenance part deals with repair and servicing.

Research and Development: This area requires the application of complex skills and expertise on research and further development to exploit robots functioning in particular areas of robotics such as underwater robotics, robots for medical purposes, GPS navigation or warfare robots and robots for industrial applications. This involves research on deriving better algorithms, software application development, interface driver development and RTOS customisation in parallel with hardware used.

Taking up academics is also a paying option, which involves working on different research projects (sponsored and non-sponsored) at university labs. Also, the scope for robotics engineers is brighter abroad since India is yet to match the pace of technological advancement.

Institutes offering courses on Robotics and Automation		
■ All Indian Institutes of Technology	■ Veermata Jijabai Technological Institute, Mumbai	
■ NIT-Calicut, Warangal, Surat, Bhopal, Rourkela, Tiruchirapalli, Durgapur	■ Jadavpur University, Kolkata	
■ IIIT-Hyderabad, Bangalore, Bhubaneswar, Allahabad, Calcutta	■ M.S. University, Baroda	
■ National Institute of Technology, the University of Hyderabad	■ NIRMA Institute of Technology	
■ Kalinga Institute of Information Technology	■ Jaypee Institute of Information Technology, Noida	
■ IISc, Bangalore	■ Delhi College of Engineering, Delhi	
■ International Institute of Technology, Pune	■ College of Engineering, Guindy, Anna University	
■ Birla Institute of Technology and Science, Pilani		

**Indicative list. Check regional technical institutes for Robotics and A.I. as electives in final year of graduation.*